sMDD vs MDD — DD Python, paired rows (same instance, DD type, width and reduction)
MDD: results_python_26_10_2026_16-30-00.csv | sMDD: results_python_semi_semi_20_10_2026_17-14-45.csv
Columns are means over the paired rows; 'ratio' is the geometric mean of sMDD/MDD per row. Time = construction + reduction; Solve = longest/shortest path over the built diagram.
Bound means skip rows where either value is ±inf (empty restricted / unreachable terminal), counted in 'non-fin.'. 'sMDD +/=/-': rows where the sMDD bound is better / equal / worse.

Problem	DD	Red.	N	Time MDD	Time sMDD	ratio	Solve MDD	Solve sMDD	ratio	Nodes MDD	Nodes sMDD	ratio	Arcs MDD	Arcs sMDD	ratio	Bound MDD	Bound sMDD	non-fin.	sMDD +/=/-
Knapsack	Exact	no	21	7.55	14.65	1.69×	2.67	3.30	1.52×	906,750	877,435	0.54×	1.69e+06	5.54e+06	2.25×	7,599	7,599	0	0/21/0
	Exact	yes	21	9.49	21.03	1.90×	2.55	4.63	2.03×	872,545	872,545	1.00×	1.64e+06	5.51e+06	2.71×	7,599	7,599	0	0/21/0
	Restricted	no	156	13.26	18.67	1.48×	4.34	5.06	1.46×	1.81e+06	1.78e+06	0.61×	3.05e+06	5.41e+06	1.81×	15,583	15,583	0	0/156/0
	Restricted	yes	156	15.83	24.71	1.67×	4.22	5.44	1.68×	1.77e+06	1.77e+06	0.97×	3e+06	5.39e+06	2.05×	15,583	15,583	0	0/156/0
	RelaxPriority	no	156	15.49	17.67	1.30×	4.12	4.25	1.33×	1.81e+06	1.79e+06	0.62×	3.06e+06	4.67e+06	1.65×	591,205	590,478	0	55/101/0
	RelaxPriority	yes	156	18.27	22.35	1.44×	4.15	5.06	1.51×	1.77e+06	1.75e+06	0.95×	3e+06	4.59e+06	1.80×	591,205	590,478	0	55/101/0
Independent Set	Exact	no	12	0.4952	5.35	1.93×	0.1442	0.1199	0.83×	88,554	17,328	0.62×	112,731	363,090	1.16×	30.83	30.83	0	0/12/0
	Exact	yes	12	0.5884	5.69	1.80×	0.1440	0.1203	0.81×	88,554	17,328	0.62×	112,731	363,090	1.16×	30.83	30.83	0	0/12/0
	Restricted	no	334	10.91	28.95	2.33×	1.26	0.0680	0.00×	880,840	19,126	0.00×	918,923	160,378	1.06×	41.08	41.25	237	4/89/241
	Restricted	yes	334	11.78	28.90	2.15×	1.25	0.0312	0.00×	876,584	15,375	0.00×	914,016	54,372	0.60×	41.08	41.25	237	4/89/241
	RelaxPriority	no	301	9.18	39.04	2.67×	1.18	1.28	1.05×	843,945	870,929	0.95×	912,748	1.1e+06	1.14×	288	305	0	84/61/156
	RelaxPriority	yes	301	10.16	40.25	2.51×	1.12	0.4575	0.45×	694,886	278,086	0.44×	754,561	356,827	0.52×	288	305	0	84/61/156
Set Cover	Exact	no	178	0.3525	5.48	2.34×	0.0527	1.04	3.97×	39,335	34,222	0.99×	76,791	2.29e+06	6.16×	3.61	3.61	0	0/178/0
	Exact	yes	177	0.3805	6.83	3.23×	0.0433	0.4051	3.40×	32,974	31,617	1.00×	64,926	1.19e+06	6.03×	3.61	3.61	0	0/177/0
	Restricted	no	1796	7.71	14.55	3.17×	0.1915	2.45	5.71×	128,228	155,632	1.11×	249,233	6.43e+06	9.62×	8.65	20.08	0	0/992/804
	Restricted	yes	1771	8.06	17.92	4.16×	0.1742	0.5798	2.71×	116,633	86,624	0.97×	228,603	1.6e+06	4.45×	8.66	20.17	0	0/990/781
	RelaxPriority	no	1802	7.81	15.22	3.07×	0.1927	2.45	5.57×	132,591	167,814	1.14×	263,179	6.65e+06	9.33×	2.65	2.02	0	1/1038/763
	RelaxPriority	yes	1773	7.83	18.43	4.02×	0.1796	0.5268	2.52×	116,372	79,486	0.97×	231,685	1.44e+06	4.10×	2.66	2.05	0	1/1033/739
SOC Knapsack	Exact	no	80	1.92	15.04	3.57×	0.4423	1.98	2.53×	211,490	6,746	0.08×	353,702	5.5e+06	6.21×	774	774	0	0/80/0
	Exact	yes	78	1.98	14.99	4.37×	0.0051	0.8167	23.31×	5,164	4,616	0.97×	9,922	2.32e+06	34.11×	765	765	0	0/78/0
	Restricted	no	530	0.2397	8.38	7.58×	0.0436	0.7296	1.32×	32,318	3,916	0.05×	39,681	2.11e+06	11.42×	677	753	89	261/179/90
	Restricted	yes	527	0.2636	9.41	8.95×	0.0011	0.3405	7.97×	1,574	2,856	0.47×	2,697	990,738	39.04×	676	751	89	258/179/90
	RelaxPriority	no	530	1.15	8.59	4.15×	0.0538	0.6893	3.36×	35,048	8,224	0.24×	51,239	2.12e+06	6.60×	892	866	0	248/280/2
	RelaxPriority	yes	526	1.11	9.01	5.11×	0.0015	0.3011	11.22×	1,834	2,909	1.06×	3,331	904,976	15.28×	890	864	0	245/279/2
Sequencing	Restricted	no	140	6.63	50.34	5.04×	0.4902	1.51	2.96×	49,663	47,911	0.96×	251,627	252,658	1.01×	630,022	612,286	0	137/3/0
	Restricted	yes	140	6.79	50.22	4.91×	0.1712	1.10	6.08×	11,933	29,939	2.41×	49,676	148,630	2.86×	630,022	612,286	0	137/3/0
	RelaxPriority	no	86	4.51	39.71	8.84×	1.63	50.60	28.84×	27,134	25,928	0.95×	1e+06	1.31e+07	12.64×	97,836	97,563	0	2/14/70
	RelaxPriority	yes	97	5.60	77.41	12.20×	0.5432	15.19	25.47×	2,775	16,391	5.97×	88,521	1.78e+06	17.45×	89,059	88,914	0	2/44/51